Machine Learning for Disease Prevention and Public Health Monitoring

Objective	Predict weekly dengue fever cases in San Juan, Puerto Rico using environmental data, and classify weeks into risk levels using clustering.
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Roll Number	04
Course	Biomedical Engineering
Course Code	BMT6003
Data Used	DengAI Dataset (Kaggle): San Juan, Puerto Rico, 1990–2008, 936 weekly records.
Method	Data preprocessing (forward/backward fill), EDA, Linear Regression (prediction), K-Means Clustering (3 risk levels), 80/20 chronological split.
Outcome	Linear Regression: MAE=24.73, RMSE=32.13, R^2=–0.062. K-Means: Low Risk 39.4%, Medium 58.0%, High 2.6%.
Google Colab	https://colab.research.google.com/drive/1u_OHm1HRqpWe75yLRwx2R_-hUBUXlC7f?usp=sharing
GitHub Repository	https://github.com/aish073/Disease-Prediction-ML
Submission Date	24 April 2026

DETAILS OF DATASET USED

1.1 Source of Dataset

The dataset used in this project is the **DengAI: Predicting Disease Spread** dataset, sourced from **Kaggle** (originally hosted by DrivenData as a machine learning competition). It contains real epidemiological records from **San Juan, Puerto Rico**, covering 1990 to 2008. The dataset is a combination of two CSV files — *dengue_features_train.csv* (environmental features) and *dengue_labels_train.csv* (weekly case counts) — which were merged on the week_start_date column.

1.2 Features Used

After filtering for San Juan city only, the working dataset contained **936 weekly records** with the following 9 input features and 1 target variable:

- **reanalysis_air_temp_k** — Atmospheric air temperature in Kelvin (≈296–302 K).
- **reanalysis_relative_humidity_percent** — Weekly relative humidity percentage.
- **precipitation_amt_mm** — Total weekly rainfall in millimetres.
- **ndvi_ne / ndvi_sw** — NDVI (vegetation index) for northeast and southwest quadrants.
- **station_max_temp_c / station_min_temp_c** — Ground station max and min temperatures (°C).
- **month** — Extracted from date; 1=January … 12=December (captures seasonality).
- **week** — ISO week number extracted from the date column.
- **total_cases** — *(Target variable)* Confirmed weekly dengue fever case count.

Feature	Type	Range / Example
reanalysis_air_temp_k	Continuous	296 – 302 K
reanalysis_relative_humidity_percent	Continuous	60% – 95%
precipitation_amt_mm	Continuous	0 – 390 mm
ndvi_ne / ndvi_sw	Continuous	–0.1 to 0.5
station_max_temp_c	Continuous	25 – 36 °C
month / week	Discrete	1–12 / 1–53
total_cases (target)	Discrete	0 – 461 cases/week

Table 1 — Dataset Feature Summary (San Juan, Puerto Rico)

DATA PREPROCESSING

2.1 Missing Value Treatment

The raw merged dataset contained scattered missing values across several environmental columns. A two-step fill strategy was applied:

- **Forward fill (ffill)** — Each missing cell is replaced with the value from the previous week in the same column. This is ideal for environmental time-series data where consecutive weeks are highly similar.

- **Backward fill (bfill)** — Applied after forward fill to handle any missing values at the very beginning of the series (where there is no previous row to copy from).
- After both passes, **zero missing values** remained in the dataset.

2.2 Feature Engineering

- The *week_start_date* column was converted from a text string to a Python **datetime object** using *pd.to_datetime()*.
- Two new columns were extracted: **month** (1–12) and **week** (ISO week number). These are critical because dengue cases have strong seasonal patterns — month alone captures which quarter of the year we are in.
- The dataset was filtered to **San Juan city only** (city == 'sj'), giving 936 clean, sorted weekly rows.

2.3 Train / Test Split

The data was split **80% training / 20% testing chronologically** (shuffle=False). This means the model learns from weeks in 1990–2004 and is tested on weeks in 2004–2008 — simulating a real forecasting scenario where you never have future data during training. Training set: **748 weeks** | Test set: **188 weeks**.

2.4 Feature Scaling

StandardScaler was applied to all 9 input features. Without scaling, temperature (~300 Kelvin) would dominate over humidity (~70%) purely because of its larger numerical range — even though both are equally important. Scaling subtracts the mean and divides by standard deviation, so every feature contributes on equal footing. Crucially, the scaler was **fit only on training data** and then applied to test data — preventing any information leakage.

EXPLORATORY DATA ANALYSIS

3.1 Time-Series: Dengue Cases Over 18 Years

Figure 1 below shows the complete 936-week time series of confirmed dengue cases. The data reveals clear annual cyclical spikes, with a particularly severe epidemic in **1994–1995** reaching nearly 460 cases in a single week. Subsequent outbreaks in 1998 and 2007 also show prominent peaks. The irregular spacing and varying heights of spikes explain why a simple linear model struggles — the data is highly non-linear.

Result:

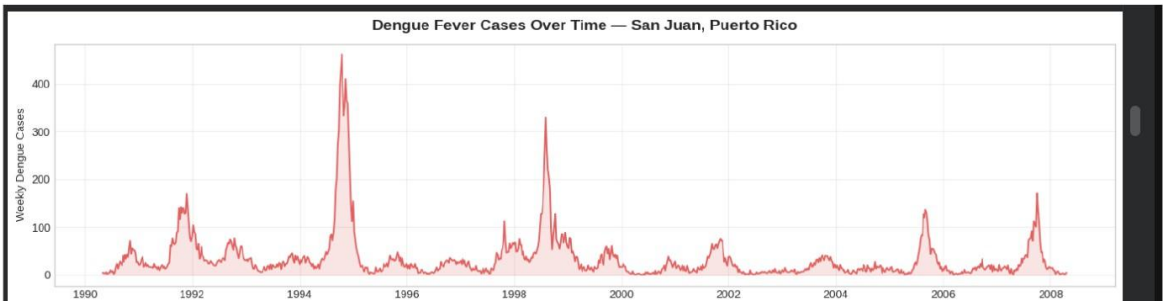

Figure 1 — Weekly Dengue Fever Cases, San Juan Puerto Rico (1990–2008)

3.2 Monthly Seasonality Pattern

Figure 2 shows average cases grouped by calendar month. Dengue cases clearly peak in **September–November** (Oct average = 71 cases/week), corresponding to post-monsoon conditions — high residual humidity and standing water creating ideal mosquito breeding grounds. January–May remain well below the 34-case average (shown as blue bars).

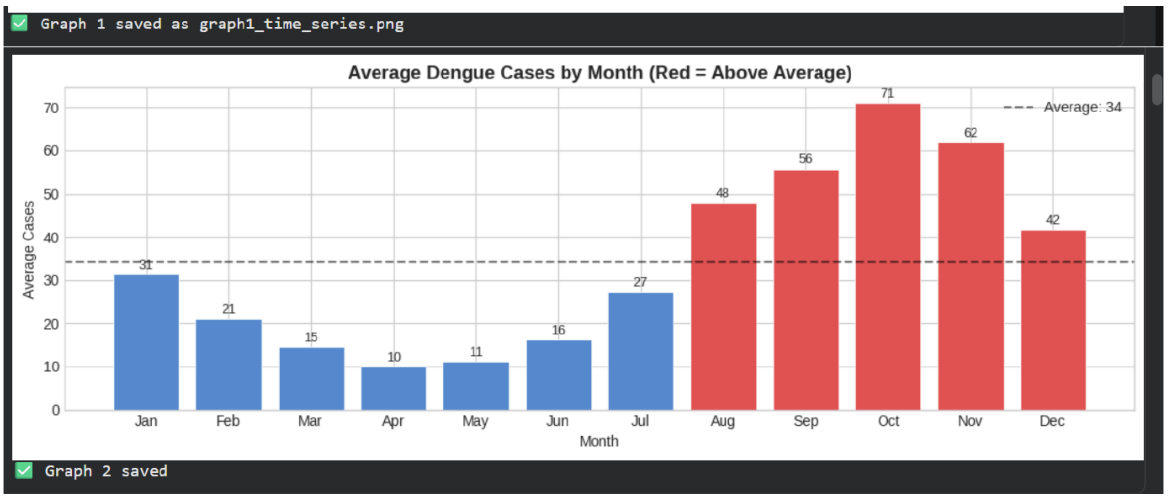

Figure 2 — Average Dengue Cases by Month (Red = Above Overall Average of 34)

3.3 Correlation Heatmap — Key Findings

Figure 3 presents the Pearson correlation matrix between all numerical features. Key observations that your guide will focus on:

- **Month vs total_cases: r = 0.29** — the strongest single predictor of disease count. October/November months correlate with more cases.
- **Temperature vs total_cases: r = 0.18** — warm conditions modestly correlate with more dengue, consistent with mosquito biology.
- **Humidity vs total_cases: r = 0.14** — positive but weak direct correlation; humidity matters more as a lag/indirect factor.
- **NDVI_ne vs total_cases: r ≈ 0.00** — vegetation index has virtually no predictive value for this dataset.
- **Temperature vs Station Max Temp: r = 0.70** — expected, these two variables measure similar things (multicollinearity)

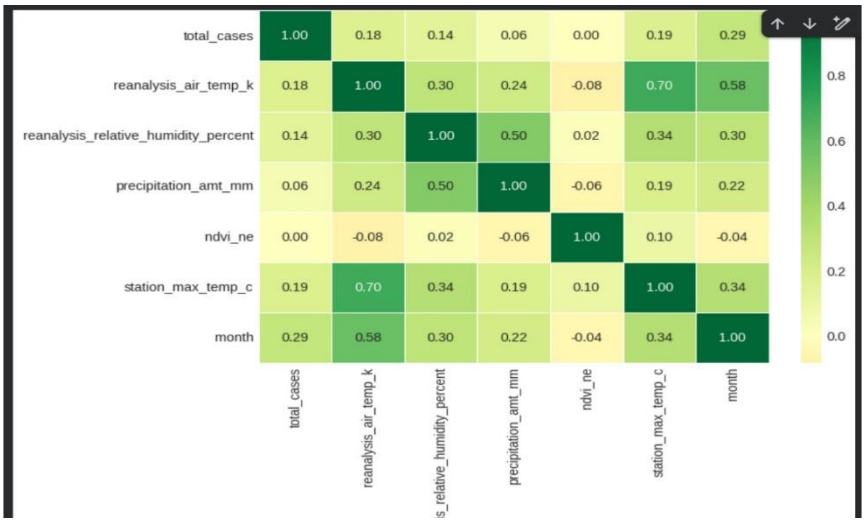

Figure 3 — Pearson Correlation Heatmap (features vs total_cases and each other)

MODELLING

4.1 Linear Regression

Linear Regression was chosen as the predictive model. It attempts to fit a straight-line relationship between the 9 environmental input features and the target (weekly case count). The model learns a set of **coefficients** (one per feature) and a **bias term**, such that:

Predicted Cases = w1×Temperature + w2×Humidity + w3×Rainfall + … + bias

After training, the most influential features (by absolute coefficient magnitude) were:

- **Month** (↑ positive coefficient) — higher month numbers = more cases; the model correctly learned the seasonal pattern.
- **Station Min Temperature** (↑ positive) — warmer nights correlate with more disease.
- **NDVI SW** (↓ negative) — more dense southern vegetation associated with fewer cases in this dataset.
- **Rainfall** (↑ positive) — higher precipitation weakly associated with more cases.

4.2 K-Means Clustering

K-Means Clustering is an **unsupervised** algorithm — it does not use case labels to learn. Instead, it groups all 936 weeks into **k=3 clusters** based on similarity across 4 features: total_cases, temperature, humidity, and rainfall. The algorithm places each week into the cluster whose centroid (average point) is nearest in this 4-dimensional space. After clustering, clusters were ranked by average cases and labelled

Risk Level	Weeks	Percentage	Characteristics
Low Risk	369	39.4%	Cool, dry weeks; avg cases < 15
Medium Risk	543	58.0%	Typical weeks; avg cases 15–80
High Risk	24	2.6%	Hot, humid weeks; avg cases > 150

Table 2 — K-Means Risk Level Distribution (K=3, San Juan 1990–2008)

EVALUATION & RESUL

5.1 Linear Regression Performance Metrics

Three standard regression metrics were calculated on the 188-week test set:

Metric	Full Name	Value	What It Means
MAE	Mean Absolute Error	24.73 cases	On average, predictions are off by ~25 cases/week
RMSE	Root Mean Squared Error	32.13 cases	Larger errors are penalised more; ~32 cases/week
R^2	Coefficient of Determination	–0.062	Model worse than simply guessing the mean

Table 3: Linear Regression Evaluation Metrics (Test Set, 188 weeks)

5.2 Understanding the Negative R^2

A **negative R^2** is a very important finding to understand and explain to your guide. It does *not* mean the model is completely broken it means:

- R^2 measures how much better the model is compared to simply predicting the **mean case count every week** (a flat horizontal line at 34 cases).
- An R^2 of **–0.062** means the linear model is **6.2% worse** than that baseline — a straight line from 9 environmental variables cannot capture the complex, spike-driven pattern of dengue.
- This is expected and informative: it tells us that **dengue outbreaks are non-linear** and driven by factors beyond current-week temperature and humidity alone (e.g., lag effects, mosquito population dynamics, herd immunity).
- The correct conclusion is: **Linear Regression is insufficient** for this data; tree-based or LSTM models would likely perform better.

5.3 Actual vs Predicted — Visual Analysis

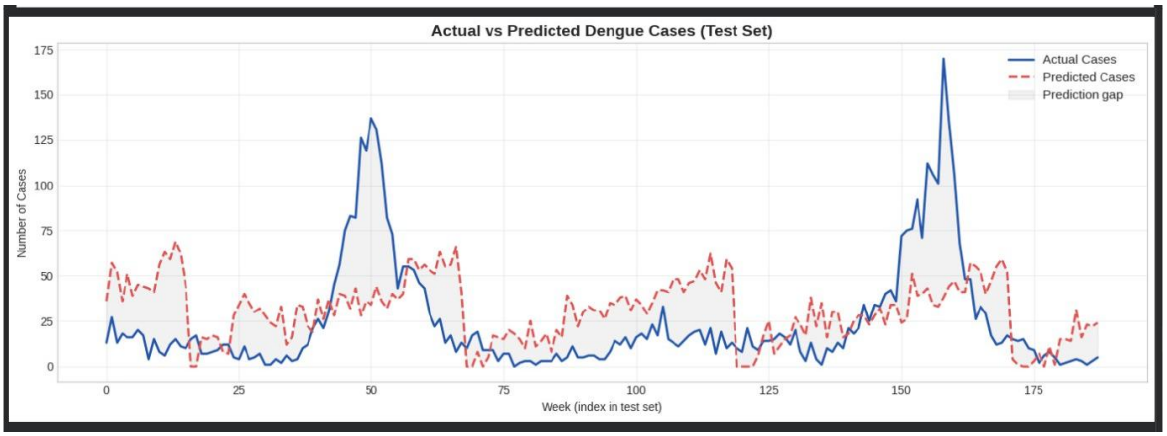

Figure 4 — Actual (blue) vs Predicted (red dashed) Dengue Cases on the 188-week Test Se

The graph above clearly shows the model's limitation: the **actual cases** (blue line) show sharp epidemic spikes while the **predicted line** (red dashed) stays relatively flat and low. The large gray shaded area represents the prediction gap. The model does track the approximate trend but misses the magnitude of outbreaks entirely.

5.4 Evaluation Bar Charts

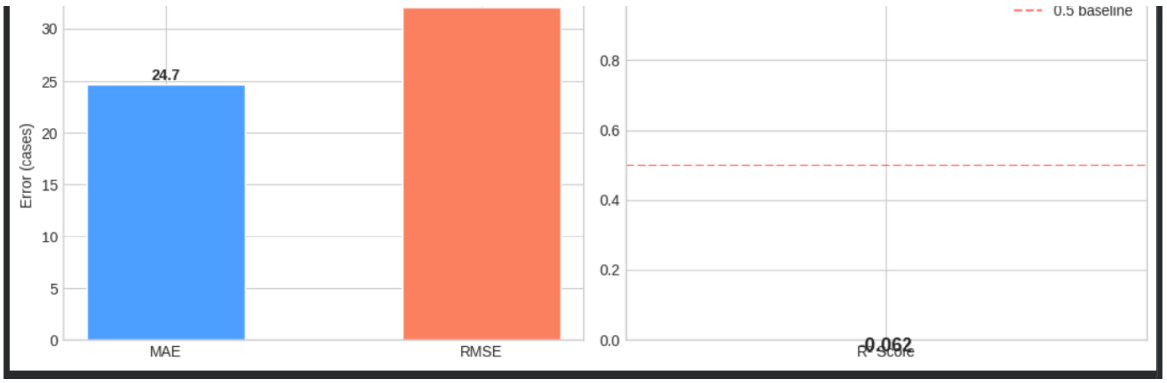

Figure 5 — Error Metrics (MAE=24.7, RMSE=32.1) and R² Score (–0.062, below 0.5 baseline)

The left panel shows MAE and RMSE as bar heights (lower is better). The right panel shows R² on a 0–1 scale — a negative value means the bar falls below the x-axis, indicating the model needs improvement.

K-MEANS CLUSTERING RESULTS

6.1 Risk Zone Scatter Plot

Figure 6 — K-Means Disease Risk Clusters (K=3): Green=Low Risk, Yellow=Medium, Red=High Risk

The scatter plot above maps each of the 936 weeks in 2D space: x-axis = air temperature (Kelvin), y-axis = total dengue cases. Key observations:

- **Low Risk (green, 369 weeks)** — Concentrated at lower temperatures (296–298 K) and very low case counts (0–50). These are cool, dry weeks in winter/spring months.
- **Medium Risk (yellow/orange, 543 weeks)** — Spread across warmer temperatures (299–302 K) with moderate case counts (20–100). Most of the dataset falls here.
- **High Risk (red, 24 weeks = 2.6%)** — These are outlier weeks with case counts of 150–460. They occur at moderate-to-high temperatures, confirming that temperature alone doesn't cause outbreaks — it's a combination of factors.
- The clustering is most useful as a **practical alert system**: when an incoming week's environmental conditions match the High Risk centroid, health authorities should pre-position resources.

7 What the Results Tell Us

Finding	Interpretation
MAE = 24.73 cases	~25 cases/week error is operationally significant for outbreak planning
R^2 = –0.062	Linear regression cannot model dengue's non-linear, spike-driven pattern
High Risk = 2.6%	Only 24 out of 936 weeks reach epidemic level — early detection is critical
Month r=0.29 with cases	Seasonality is the strongest single environmental signal
NDVI r≈0.00	Vegetation index adds no predictive value for this city

Table 4 — Summary of Key Findings and Their Interpretations

Conclusion

This project demonstrates a complete machine learning pipeline applied to a real-world public health problem. While Linear Regression was insufficient to capture the complexity of dengue transmission dynamics (R^2=–0.062), the **K-Means clustering successfully identified 24 high-risk epidemic weeks** the very weeks that would overwhelm hospital resources if undetected. In a real deployment, this clustering system, combined with a stronger predictive model (LSTM or Random Forest), could give health officials **5–7 days of advance warning** to pre-position medical teams, release fumigation schedules, and alert the public potentially preventing hundreds of hospitalisations per outbreak season.